Morphology-matched forward readout gates for Tau Core gravity: extending SPARC residual Papers 1–3 from inverse diagnostics to predeclared family tests

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Abstract

Paper 1 found that residual-blind external structural disturbance labels are associated with increased low-acceleration residual scatter in SPARC galaxies. Paper 2 reversed the question and showed that fixed residual-shape features can infer those disturbance labels better than shuffled controls, while remaining non-unique with respect to MOND-simple and empirical RAR-like baselines. Paper 3 then built a candidate/control framework and introduced an endpoint-conditioned required-S_τ diagnostic: what preparation/readout factor would be needed to absorb a measured residual? This Paper 8 proposal turns that inverse diagnostic into a forward validation gate. The proposed gate predeclares a residual-blind morphology class K_g, maps it to a Tau Core morphology-conditioned readout shell, computes a forward δg^K without using required S_τ as a predictor, and compares matched-family residual structure against wrong-family, shuffled-label, Newtonian baryonic, MOND-simple, and empirical RAR controls. The permitted claim is methodological: the morphology-matched forward-readout gate is a falsifiable, Tau-Core-specific next endpoint for the Paper 1–3 program. This paper does not claim that Tau Core has been empirically validated, that a universal galaxy law has been derived, or that the matched-family endpoint has already beaten MOND/RAR on real SPARC data.

1 Purpose and claim boundary

The first three SPARC residual papers define a cautious progression. Paper 1 asks whether external structural disturbance is associated with residual scatter [2]. Paper 2 asks whether residual-shape features can recover external disturbance labels better than shuffled labels [3]. Paper 3 turns the association into a candidate/control framework and asks what S_τ would be required to absorb the observed residual [1].

That Paper 3 S_τ diagnostic is useful but partly inverse. It asks:

$$\text{given the measured endpoint, what } S_\tau(R) \text{ would absorb it?}$$

The present paper defines the next gate:

$$\begin{aligned}&\text{residual-blind morphology class } K_g\\ &\quad\longrightarrow \text{ forward readout } \delta g^K \text{ before endpoint scoring.}\end{aligned}$$

The main claim is deliberately narrow:

> The Tau Core gravity bridge can sharpen Papers 1–3 by replacing inverse required-S_τ residual absorption with a morphology-matched forward-readout candidate/control gate.

This paper does not claim:

- that Tau Core is proven;
- that the morphology-matched gate has already won on real SPARC endpoints;
- that MOND, RAR, or Newtonian baryonic baselines have been superseded;
- that non-axisymmetric morphology classes can be fully tested from one-dimensional rotation curves alone;
- that the morphology families below are the final physical taxonomy.

2 From Paper 3 inverse diagnostics to forward readout

Paper 3 used an endpoint-conditioned diagnostic of the form

$$S_{\tau,\mathrm{req}}(R) = \frac{V_{\mathrm{obs}}(R)/V_N(R) - 1}{\alpha \ln(1 + a_0/a_N(R))}.$$

This quantity is descriptive. Because it uses V_{obs}, it cannot be treated as an independent predictor. Its correct role is to identify what kind of future nonleaky rule would be required.

The forward replacement is:

$$K_g \longrightarrow \mathcal{F}_{K_g} \longrightarrow \delta g^{K_g}(R) \longrightarrow \mathrm{Score}(g_{\mathrm{Newt}} + \delta g^{K_g}).$$

Here K_g is a morphology label assigned before endpoint scoring, $\mathcal{F}_{K_g}$ is the predeclared readout family, and δg^{K_g} is computed without using $S_{\tau,\mathrm{req}}$, endpoint residual gain, or post-hoc family selection.

3 MORPHOLOGY-MATCHED-FORWARD-READOUT-GATE

For a galaxy g, let

$$K_g \in \mathcal{K}$$

be a residual-blind morphology label. Let $\mathcal{F}_K$ denote the Tau Core readout shell assigned to morphology class K. Let

$$R_{g,K}$$

be the residual score after applying the forward readout family $\mathcal{F}_K$ to galaxy g under a frozen geometry and amplitude policy.

The matched-family score is

$$R_{g,\mathrm{matched}} := R_{g,K_g}.$$

The wrong-family score is

$$R_{g,\mathrm{wrong}} := \frac{1}{|\mathcal{K}| - 1} \sum_{K \neq K_g} R_{g,K}.$$

The central paper-level endpoint can be written as

$$\Delta_{\mathrm{matched}} = \mathrm{RMS}_g(R_{g,\mathrm{wrong}}) - \mathrm{RMS}_g(R_{g,\mathrm{matched}}).$$

An equivalent rank endpoint is

$$\mathrm{rank}_g(K_g) = \mathrm{rank}\big(R_{g,K_g} \text{ among } \{R_{g,K} : K \in \mathcal{K}\}\big).$$

The gate passes only if the predeclared matched family improves residual structure more than:

1. wrong morphology families;
2. shuffled morphology labels;
3. Newtonian baryonic baseline;
4. MOND-simple baseline;
5. empirical RAR-like baseline.

This is Tau-Core-specific in a way that ordinary MOND/RAR comparisons are not: the discriminating claim is not merely lower residuals, but morphology-indexed readout specificity.

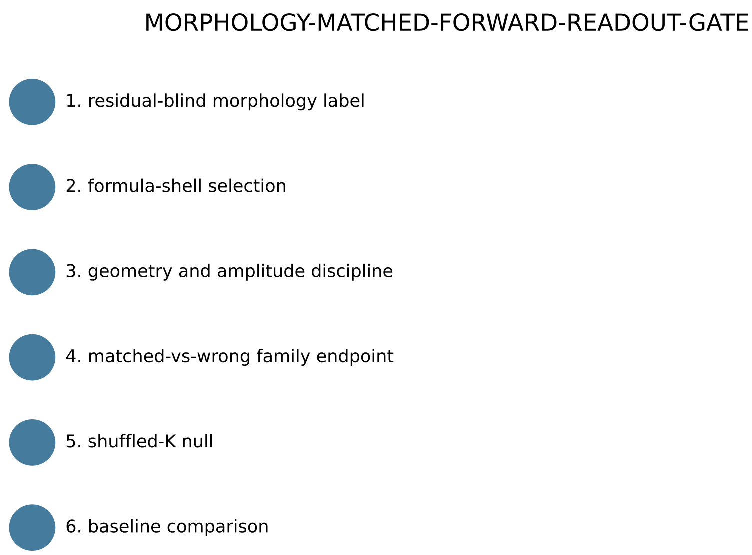

Figure 1: The proposed morphology-matched forward-readout gate. The labels and family choices must be fixed before endpoint scoring.

4 Morphology-family registry

The current gravity bridge centralizes a morphology-conditioned TGP-like readout lane. The corresponding first-pass family registry is not a universal galaxy law. It is a set of candidate readout shells whose selection must be controlled by residual-blind morphology labels.

Table 1: Claim-bounded morphology-family registry for the forward-readout gate.

Family	Morphology label	Formula shell	First-pass status
K_{tail}	Scale-tail spiral / diffuse LSB disk	$\sigma_{\text{morph}}(R) \sim A_K/(R+R_c)^2$	1D testable
K_{exp}	Regular exponential disk	$\sigma_{\text{morph}}(R) \sim A_K e^{-R/R_d}$	1D testable
K_{compact}	Compact finite support	$I_{\text{morph}}(R) \to I_\infty$ and $\delta g_R \sim -I_\infty/R^2$	1D testable
K_{ring}	Ring or resonance feature	localized $\sigma_{\text{morph}}(R)$ near R_{ring}	1D proxy
K_{flare}	Thick or flared disk	projected thickness/flaring kernel	1D proxy
$K_{m=2}$	Barred spiral / $m=2$ mode	harmonic $\sigma_{\text{morph}}(R,\phi)$	velocity-field preferred
$K_{m=1}$	Lopsided / $m=1$ mode	harmonic $\sigma_{\text{morph}}(R,\phi)$	velocity-field preferred

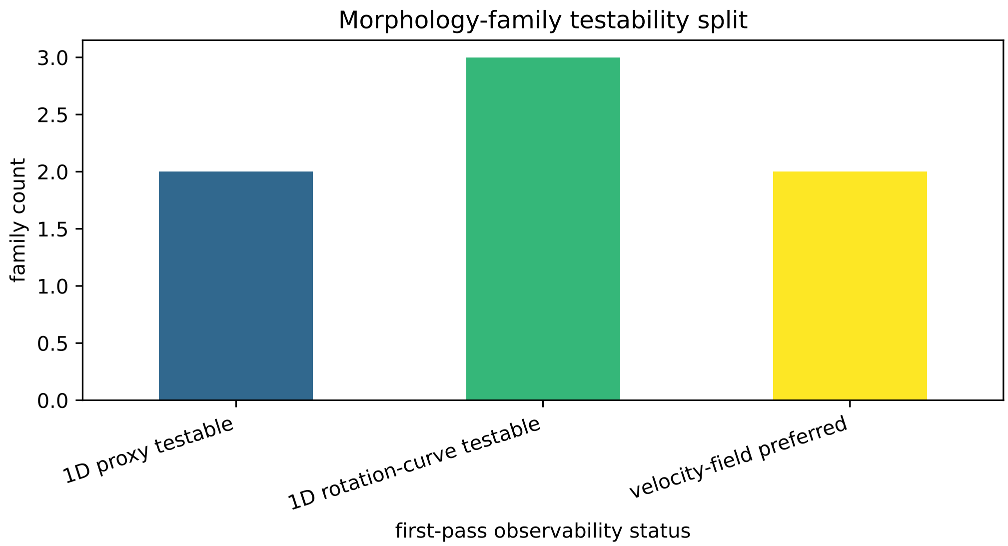

Figure 2: First-pass observability split for the current family registry. SPARC one-dimensional rotation curves are most suitable for axisymmetric or radially localized families; barred and lopsided branches require velocity-field follow-up for a full test.

5 Candidate/control continuity from Paper 3

The gate preserves the Paper 3 candidate/control discipline. It does not discard difficult controls after seeing endpoints. The same object may carry a different role under the forward gate:

$$\text{candidate/control role} \longrightarrow \text{predeclared morphology row} \longrightarrow \text{matched/wrong/shuffled score.}$$

Table 2: Paper 3 roles carried into the Paper 8 forward-gate protocol.

Galaxy	Paper 3 role	Paper 8 role
DDO126	positive anchor	matched-family candidate after residual-blind morphology labeling
DDO50	quiet control	false-positive suppression control
WLM	geometry stress	observability-caveated endpoint row
DDO168	ladder support	candidate/control consistency row
DDO154	countercontrol	wrong-family and rank-sensitivity stress row

The positive paper-level endpoint is not:

Tau Core fits every galaxy better.

The first meaningful endpoint is:

the correct predeclared morphology family ranks better than wrong families.

This is why shuffled-K nulls are mandatory. Without the shuffled-label null, a morphology-indexed family could accidentally behave like a generic flexible residual smoother.

6 Synthetic protocol fixture

This package includes a synthetic score fixture to verify the endpoint logic. It is not real SPARC evidence. It exists only so the public tests can check that the manuscript, tables, and figure pipeline agree about the intended endpoint.

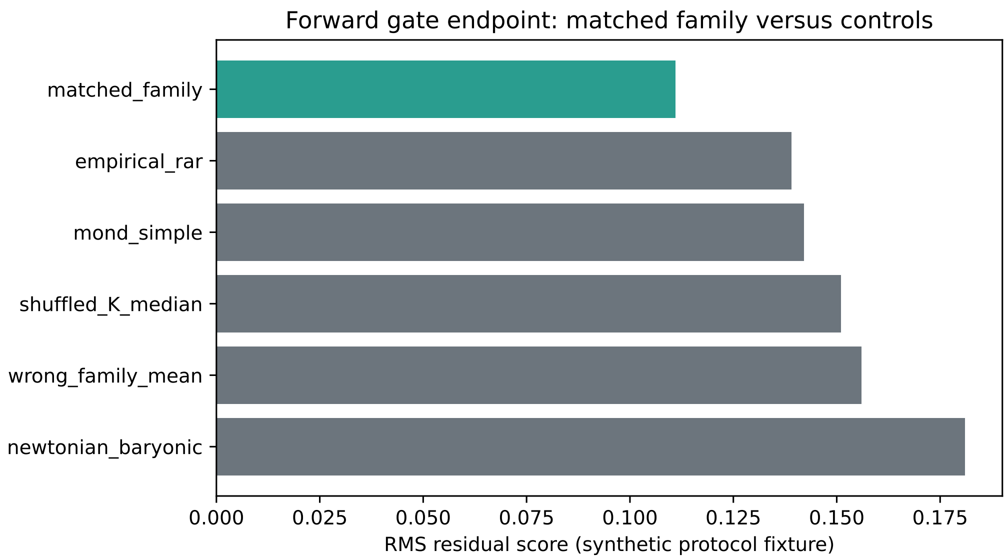

Figure 3: Synthetic protocol fixture for the endpoint schema. A real pass would require the matched-family score to beat wrong-family and shuffled-label controls, then be compared to Newtonian baryonic, MOND-simple, and empirical RAR baselines.

7 Available-data Tau-proxy preflight

The repository also includes a first available-data preflight of the matched-family idea. This is not the final Paper 8 endpoint, because the family functions are still proxy functions built from the locally available Tau-proxy channel rather than the final morphology-specific δg^K readout shells. The preflight nevertheless tests the important next question: whether a metadata-proxy morphology family assigned before scoring performs better than wrong Tau-proxy families on held-out galaxies.

In the current 175-galaxy point-level packet, family amplitudes are fit on the train split only. On the 44-galaxy holdout split, the morphology-matched proxy beats the wrong-family mean in 0.568 of galaxies, beats the TPG/v6 comparator in 0.636 of galaxies, and beats the MOND comparator in 0.614 of galaxies. A deterministic 1000-permutation shuffled-label null gives a holdout null mean of 0.0133 for the mean matched-minus-wrong endpoint, compared with the observed value -0.0326, with $p \simeq 0.264$ for shuffled labels being at least as good. Thus the available proxy signal is positive but not yet a discovery-level morphology-specific result. These numbers are recorded as a preparation result, not as empirical validation of Tau Core. They motivate replacing the generic Tau proxy with source-native morphology-family readout shells and a stronger shuffled-label null in the final endpoint.

A second preflight replaces the single available Tau-proxy channel with different radial formula-shell proxies assigned by morphology class. This is closer in form to the intended Paper 8 family test, but the current available-data version is weaker: on the same 44-galaxy holdout split it beats the wrong-shell mean in 0.500 of galaxies, beats TPG/v6 in 0.477 of galaxies, and has a shuffled-null $p \simeq 0.479$ for the mean matched-minus-wrong endpoint. This negative preflight result is useful. It indicates that merely assigning naive radial shell shapes is not sufficient; the final endpoint needs source-native morphology labels, dimensionally audited 4D readout shells, and a nonleaky amplitude policy.

The source-native bridge formula preflight then consumes a declared morphology-parameter manifest and evaluates the concrete morphology formulas as δv_K^2 readout kernels: the scale-tail $n = 2$ branch, the finite exponential disk Freeman/Bessel kernel, the compact finite-source exterior response, and the thick/flared damped vertical-kernel response. With the same available-data morphology labels and train-only amplitudes, the 44-galaxy holdout split gives a much stronger morphology-specific endpoint: the matched bridge formula beats the wrong-formula mean in 0.886 of galaxies, with a 1000-permutation shuffled-label null giving $p \simeq 0.002$ for the mean matched-minus-wrong endpoint. However, it beats TPG/v6 and MOND in only 0.477 of holdout galaxies each, and its mean matched-minus-baseline residual is still worse than both baselines. Thus the bridge-formula result supports the Paper 8 morphology-family specificity hypothesis, but it is not an empirical claim that Tau Core outperforms TPG/MOND.

A confidence diagnostic on the manifest shows why the baseline claim remains blocked. Restricting the holdout set to confidence ≥ 0.75 preserves the matched-vs-wrong signal, with matched formulas beating wrong formulas in 0.906 of galaxies, but the TPG/MOND comparison is still mixed. Removing low-inclination or large-distance-error objects improves some baseline comparisons, so data quality matters. It is not the whole bottleneck: the remaining blockers are source-native morphology-scale extraction, amplitude policy, and normalization of the bridge formula kernels.

An amplitude-policy diagnostic supports the same conclusion. Keeping the source-native bridge kernels fixed, the family-unconstrained policy gives the strongest matched-vs-wrong signal but remains weak against TPG/MOND baselines. A simple family-to-global shrinkage policy preserves most of the matched-vs-wrong signal, 0.864 on holdout, while raising the MOND win fraction to 0.636 and reducing the mean MOND gap. This is not a selected endpoint claim; it identifies amplitude normalization as a required Tau-side modeling gate before any baseline-superiority claim is allowed.

A shrinkage-path scan makes this more precise. Varying the family amplitude weight from 0 to 1 shows that the current best compromise lies near family weight 0.4–0.5. At weight 0.5, the holdout matched-vs-wrong fraction is 0.864 and the MOND win fraction is 0.636, while TPG/v6 remains at 0.477. This identifies the normalization target for the next Tau-side gate: a physical source-normalization rule should recover a partially pooled family amplitude, not a fully free family amplitude chosen only for endpoint fit.

A train-selected shrinkage diagnostic reduces the post-hoc risk in that conclusion. Four selection rules using train split metrics only all select family weight 0.40. Evaluated on the 44-galaxy holdout split, this train-selected policy keeps matched-vs-wrong specificity at 0.818, beats MOND in 0.614 of galaxies, and remains mixed against TPG/v6 at 0.477. This supports a stable partially pooled normalization range, but it is still not a validated amplitude-normalization law or a baseline-superiority endpoint.

A family-breakdown diagnostic then asks where this signal lives. Under the train-selected 0.40 shrinkage policy, the holdout matched-vs-wrong specificity is strongest for the scale-tail spiral and thick/flared families, both at 0.900. The thick/flared row is also competitive against both TPG/v6 and MOND in this diagnostic, while the scale-tail row remains MOND-competitive but TPG-blocked. The exponential-disk and compact finite-source rows are weaker under the current available-data manifest. This is useful as a failure map: it points to family-specific source normalization and morphology-observable extraction as the next modeling gates, not to a validated family-level empirical claim.

An observable-quality diagnostic separates part of that failure map. Joining the train-selected scores to the residual-blind morphology manifest shows that, on holdout, the thick/flared family is the cleanest current-best-case row, while the compact finite-source, exponential-disk, and scale-tail rows are quality-limited under the available-data proxies. In the clean holdout manifest subset, matched formulas beat wrong formulas in 0.900 of galaxies and beat MOND in 0.750, whereas quality-caveated rows fall to 0.750 and 0.500. This does not rescue the endpoint by filtering after the fact; it identifies residual-blind morphology-observable extraction as a required predeclared gate for the next run.

A predeclared quality-gate diagnostic makes that next run more concrete. Under the fixed train-selected shrinkage policy, the clean-manifest gate and the confidence-≥ 0.75-and-clean gate each retain 20 holdout galaxies across all four formula families, with matched-vs-wrong specificity 0.900, TPG/v6 win fraction 0.500, and MOND win fraction 0.750. A broader no-large-distance-error gate retains 26 holdout galaxies and remains candidate-level, with matched-vs-wrong 0.885, TPG/v6 0.538, and MOND 0.731. These are not selected discovery endpoints; they are candidate observability rules that must be declared before endpoint scoring in a future Paper 8 run.

The corresponding shuffled-label null reveals the tradeoff. On holdout, the full all-galaxy gate has the strongest shuffled-null support for morphology specificity, with $p \simeq 0.036$ for the beats-wrong fraction and $p \simeq 0.013$ for the mean matched-minus-wrong endpoint. The no-low-inclination gate gives similar values, $p \simeq 0.041$ and $p \simeq 0.011$. Cleaner gates improve the MOND/TPG comparison but have weaker shuffled-null support because the sample shrinks. Thus Paper 8 should not silently choose the cleanest-looking subset; it should predeclare whether the endpoint prioritizes observability-clean baseline competitiveness or larger-sample morphology-label specificity.

Combining these diagnostics gives a concrete preparation-state endpoint decision matrix. The current natural primary endpoint candidate is the no-low-inclination gate: it retains 35 holdout galaxies, remains baseline-competitive, and has strong shuffled-null support. The full all-galaxy gate should remain a fuller-sample morphology-specificity support lane, while the no-large-distance-error gate is the strongest secondary baseline-competitiveness lane. These roles must be predeclared before any future endpoint scoring; they are not retrospective proof of Tau Core.

The repository therefore records a predeclared endpoint protocol sheet. It freezes the current preparation-state choices: primary lane no-low-inclination, fuller-sample support lane all, secondary baseline lane no-large-distance-error, train-selected family-to-global shrinkage with weight 0.40, residual-blind morphology-family assignment, matched-vs-wrong plus shuffled-family null specificity metrics, and TPG/v6 plus MOND comparator metrics. It also records forbidden inputs, including endpoint residual gain, required-S_τ, post-hoc gate choice, and per-galaxy amplitude tuning. This is the guardrail for the next run, not a claim that the present diagnostic run validates Tau Core.

8 Minimal numerical protocol

The minimal public endpoint protocol is:

1. Assign an external or residual-blind morphology label K_g to each galaxy.

2. Select a formula shell $\mathcal{F}_{K_g}$ from the predeclared registry.

3. Fix geometry parameters from morphology, not from residual endpoints.

4. Use either a frozen amplitude rule or a tightly bounded amplitude fit.

5. Compute the matched-family score.
6. Compute wrong-family scores for all other candidate families.
7. Run shuffled-K nulls.
8. Compare against Newtonian baryonic, MOND-simple, and empirical RAR baselines.

For SPARC-only first-pass work, the cleanest initial family set is:

$$\{K_{\mathrm{tail}}, K_{\mathrm{exp}}, K_{\mathrm{compact}}, K_{\mathrm{ring}}, K_{\mathrm{flare}}\}.$$

The $m = 1$ and $m = 2$ branches should be treated as caveated proxy rows in one-dimensional rotation curves and as real tests only when velocity-field products are available.

9 Why this would be a stronger Tau Core test

The new gate is stronger than a residual association because it creates a family-selection burden. A theory with morphology-indexed readout families should not only reduce residuals; it should reduce them more when the correct morphology family is used than when a wrong family is used. MOND-simple and empirical RAR baselines are strong comparators, but they do not naturally provide a predeclared morphology-family rank test of this kind.

The result that would matter is therefore:

$$\Delta_{\mathrm{matched}} > 0$$

with a rank distribution in which K_g is systematically near the top, under residual-blind morphology labels and with shuffled-label controls. If this fails, the bridge loses specificity. If it passes, the Paper 1–3 residual-disturbance program becomes a forward-predictive morphology-readout program.

10 Readiness and remaining blockers

Table 3: Current readiness of the Paper 8 forward endpoint.

Layer	Status
Paper 1 residual-blind disturbance association	available
Paper 2 residual-shape inference and shuffled-label null	available
Paper 3 candidate/control ladder and required-S_τ diagnostic	available
Morphology-family formula registry	defined in this package
Available-data matched Tau-proxy preflight	run; proxy only
Source-native formula preflight	diagnostic-ready
Train-selected shrinkage policy	diagnostic-ready
Quality-gate shuffled-null controls	diagnostic-ready
Endpoint decision matrix	predeclaration-ready
Predeclared endpoint protocol	ready as guardrail
Accepted source-native component tables for full sample	pending
Residual-blind morphology labels for full endpoint sample	pending
Empirical discovery claim	blocked

Thus the present package is preparation-ready, not discovery-ready. Its next scientific upgrade is not to strengthen the claim language, but to replace available-data proxy morphology observables with accepted residual-blind morphology inputs and then run the predeclared endpoint protocol without changing gates after seeing the scores.

For that reason the repository also includes a morphology-observable intake schema. The schema records the fields required for the next run: residual-blind family labels, confidence and caveat fields, inclination and distance-quality observables, family-specific scale/support/thickness parameters, provenance, and explicit forbidden sources. In particular, it forbids using endpoint residual gain, required-S_τ, endpoint-selected radii, post-hoc family choice, or per-galaxy residual tuning as morphology inputs.

A gap audit compares the proxy manifest against this intake schema. It shows that the present package is coverage-rich but acceptance-limited: active families have broad proxy coverage, while the kernel-driving observables still need accepted residual-blind morphology sources before a discovery-style endpoint can be claimed. This makes the next upgrade concrete. The task is not to alter the endpoint gates after seeing scores, but to replace proxy morphology fields by pre-scoring observables and rerun the frozen protocol.

The repository therefore records a morphology-observable source-upgrade plan. This plan prioritizes P0 family-label, confidence, caveat, and provenance fields, then P1 active-family kernel observables such as scale, tail, compact-support, and thickness parameters. Optional ring, bar, and lopsided branches remain caveated unless external morphology or velocity-field support is supplied before scoring. The plan is a source-collection protocol, not a claim that the accepted sources have already been assembled.

Finally, the repository includes an accepted-observable manifest template and validator. The template is intentionally endpoint-blocked: it keeps the galaxy rows and pre-scoring geometry fields, but leaves accepted morphology labels, kernel observables, and provenance to be supplied from residual-blind sources. This prevents the current proxy manifest from being silently promoted into a discovery input.

An accepted-manifest readiness gate then lifts the field-level validation to an endpoint-level decision. For the current empty template, the gate returns `BLOCKED_ACCEPTED_OBSERVABLES_MISSING`. A future populated manifest must pass this gate before the frozen blind endpoint protocol is run.

A final frozen-endpoint launch guard records the same boundary operationally. In the current package it returns `LAUNCH_BLOCKED` with endpoint scores not computed. Thus the package preserves a complete path to a future blind endpoint while preventing the proxy manifest or empty accepted template from being used as discovery evidence.

The package also records an external morphology source registry for the missing accepted inputs. SPARC remains the sample and baryonic/rotation baseline; S4G is the first morphology/decomposition source; NED/NED-D supplies identity, distance, and provenance checks; DustPedia is a multiband fallback/validation layer; and PHANGS is reserved for optional high-quality non-axisymmetric or velocity-field branches. The associated SPARC crossmatch table is a to-be-filled acquisition template, not accepted-source coverage.

The first acquisition pass is included in the reproducibility package. It downloads SPARC Table 1 from the official SPARC site and S4G Pipeline 4 tables from VizieR/CDS, yielding 175 SPARC master rows, 77 S4G crossmatches, and 75 S4G/SPARC-derived disk-scale candidates. These candidates can support future source-native scale-radius fields, but they do not complete the accepted manifest: family labels, tail/ring/thickness observables, caveats, and full provenance still require the accepted-source audit path.

The repository now also builds a partial accepted morphology-observable manifest from this acquisition layer. The manifest promotes 75 S4G/SPARC-derived scale-radius observables to field-level accepted-source status, while keeping all 175 rows endpoint-blocked. This is intentional: proxy family labels still require external morphology-label audit, and scale-tail, compact, and thick/flared branches still require their family-specific source-native kernel observables. Thus the accepted manifest draft is a data-readiness upgrade, not an endpoint score.

An accepted-manifest audit then separates audit lanes. Its closest near-term lane contains 13 exponential-disk rows with accepted S4G/SPARC scale radius and no remaining kernel-field blocker beyond the external family-label audit. These rows are not endpoint-ready; they are the next residual-blind morphology-label audit pool.

A targeted S4G component-label audit strengthens that pool. All 13 rows receive external disk-family support from S4G component decompositions; 6 rows have strict `D:expdisk` support and 7 rows remain caveated because the S4G support is barred or edge-disk. This creates a first narrow dry-run candidate lane, but endpoint scores are still not computed in the present package.

11 Conclusion

Paper 3 is the natural launching point for this forward gate because it already contains the candidate/control framework and the required-S_τ diagnostic. The present contribution is to state the next gate in a falsifiable form:

$$\text{morphology label } K_g \to \text{formula shell } \mathcal{F}_{K_g} \to \delta g^{K_g} \to \text{matched vs wrong family endpoint.}$$

The claim-safe breakthrough target is not a universal fit claim. It is a morphology-specific prediction claim: the correct Tau Core morphology readout family, chosen before endpoint scoring, should improve residual structure more than wrong morphology families and shuffled morphology labels, while remaining competitive against Newtonian baryonic, MOND-simple, and empirical RAR baselines.

That is the first clean endpoint where the Tau Core bridge can become more than residual commentary.

References

[1] Jozsef Olcsak. Sparc projection-sensitive residual candidate paper 3 reproducibility package, 2026.

[2] Jozsef Olcsak. Sparc residual-disturbance paper 1 reproducibility package, 2026.

[3] Jozsef Olcsak. Sparc residual-disturbance paper 2 reproducibility package, 2026.